Package 'sloshR'

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Type Package

Title Read, Analyse and Visualise 'SLOSH' Storm Surge Model Runs

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Description Tools for working with historical and hypothetical storm runs ('rex' files) from the National Weather Service 'SLOSH' (Sea, Lake and Overland Surges from Hurricanes) storm surge model. Converts 'rex' files to text in batch through the 'SLOSH' Display Program's own library, then builds per-storm surge envelopes, site time series and event tables; computes storm metrics (peak, duration above thresholds, rate of rise), flood footprints, timing against tides and other flood drivers, validation statistics against observations and site-to-gauge transfer ratios; and draws maps, time series and animations. 'SLOSH' is described in Jelesnianski, Chen and Shaffer (1992) <https://repository.library.noaa.gov/view/noaa/7235>.

License Apache License (>= 2)

URL https://github.com/sachinbhere/sloshR

BugReports https://github.com/sachinbhere/sloshR/issues

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Imports data.table (>= 1.14.0),
ggplot2 (>= 3.4.0),
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scales,
stats,
tools,
utils

Suggests gifski,
knitr,
rmarkdown,
testthat (>= 3.0.0)

VignetteBuilder knitr

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SystemRequirements The SLOSH Display Program (Windows) is needed only for slosh_convert(); all other functions work on any platform.

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`slosh_animate` *Animate one storm*

Description

Draws the surge over the SLOSH grid frame by frame, with the storm track so far, the storm centre and (optionally) a site labelled with its current surge, and stitches the frames into a GIF.

Usage

```
slosh_animate(
  storm,
  txt_dir,
  file = paste0(storm, ".gif"),
  site = NULL,
  radius_km = 3,
  every_nth = 2,
  fps = 10,
  width = 900,
  height = 820,
  keep_frames = FALSE
)
```

Arguments

`storm`	Storm folder name, e.g. `"2008Ike_gl2"`.
`txt_dir`	Folder written by `slosh_convert`.
`file`	Output GIF path. Default `"<storm>.gif"` in the working directory.
`site`	Optional `list(name = , lat = , lon = )`.
`radius_km`	Radius for the site value (see `slosh_site_series`).
`every_nth`	Use every n-th frame (SLOSH frames are usually 10 min apart).
`fps`	Frames per second of the GIF.
`width, height`	Frame size in pixels.
`keep_frames`	Keep the PNG frames (in a folder next to the GIF).

Value

The GIF path, invisibly. If the **gifski** package is not installed, the PNG frames are kept and their folder is returned.

Examples

```
if (requireNamespace("gifski", quietly = TRUE)) {
  f <- file.path(tempdir(), "demo.gif")
  slosh_animate("2099Demo_tst", slosh_example_dir(), file = f,
                site = list(name = "Demo site", lat = 29.31, lon = -94.79),
                every_nth = 3, width = 500, height = 450)
}
```

`slosh_convert` *Convert SLOSH rex files to text*

Description

Runs NOAA's SLOSH Display Program library in batch to write every frame of every rex file as text: one folder per storm, each frame as `<storm><frame>.txt` (longitude, latitude, surge in feet) and `<storm><frame>.wnd` (storm centre, radius of maximum winds and wind speed).

Usage

```
slosh_convert(sdp_dir, pattern = "*", out_dir = NULL, tclkit = NULL)
```

Arguments

`sdp_dir`	Folder of the SLOSH Display Program, the one holding `bin`, `bnt` and `rexfiles`.
`pattern`	File-name pattern (without `.rex`), e.g. `"2008Ike*"`. Default `"*"` converts every rex file found in `rexfiles` and its subfolders.
`out_dir`	Where to write the text. Default `<sdp_dir>/output/rex_txt`.
`tclkit`	Path to NOAA's Tcl interpreter. Default `<sdp_dir>/bin/tclkit854.exe`.

Details

This step needs Windows and an installed SLOSH Display Program, because NOAA's library is a 32-bit Windows DLL. All other sloshR functions work on the converted text on any platform.

Value

The conversion log (see `slosh_read_log`), invisibly.

See Also

`slosh_read_log`, `slosh_list_storms`

Examples

```
## Not run:
log <- slosh_convert("C:/slosh.pkg/sloshdsp", pattern = "2008Ike*")
log

## End(Not run)
```

slosh_driver_lag *Lag between the surge peak and another flood driver*

Description

For compound-flood timing: compares the surge peak at a site with the peak of rainfall, river flow or any other series within a window around it.

Usage

```
slosh_driver_lag(series, driver, window_h = 72)
```

Arguments

series	Site time series (slosh_site_series), one or several storms.
driver	Data frame with time (POSIXct or text, UTC) and value; optionally storm to match storms by name.
window_h	Hours either side of the surge peak to search.

Value

A data.table, one row per storm: storm, surge_peak_time, surge_peak_m, driver_peak_time, driver_peak, lag_h (positive = driver peaks after the surge) and driver_at_surge_peak.

Examples

```
fr <- slosh_read_frames("2099Demo_tst", slosh_example_dir())
ts <- slosh_site_series(fr, lat = 29.31, lon = -94.79)
pk <- ts$time[which.max(ts$surge_m)]
rain <- data.frame(time = pk + seq(-48, 48) * 3600)
rain$value <- exp(-((seq(-48, 48) - 10) / 6)^2) * 40
slosh_driver_lag(ts, rain)
```

slosh_envelope *Maximum surge envelope of one storm*

Description

The highest water level each grid cell reached during the storm, and when.

Usage

```
slosh_envelope(frames)
```

Arguments

frames	Output of slosh_read_frames.

Value

A data.table, one row per cell: `storm`, `cell`, `lon`, `lat`, `max_surge_ft`, `max_surge_m`, `time_of_max` (UTC; NA when the cell stayed dry), `cell_km` and `ds_deg` (approximate cell size in km and degrees, estimated from neighbouring cells).

Examples

```
fr  <- slosh_read_frames("2099Demo_tst", slosh_example_dir())
env <- slosh_envelope(fr)
env[order(-max_surge_m)][1:5]
```

---

slosh_events *Event table: peak surge of every storm at a site*

Description

Reads each converted storm, finds its surge time series at the site and summarises it in one row. Storms run on a grid that does not reach the site get `covers_site` = `FALSE`. The same historical storm is sometimes run on more than one grid or bathymetry (e.g. `1900gl2` and `1900gls-asof1900`); repeats are flagged in `duplicate_event` so each storm counts once.

Usage

```
slosh_events(
  txt_dir,
  lat,
  lon,
  radius_km = 3,
  storms = NULL,
  save_dir = NULL,
  site_name = "site",
  verbose = TRUE
)
```

Arguments

`txt_dir`	Folder written by `slosh_convert`.
`lat, lon`	Site latitude and longitude (decimal degrees, west negative).
`radius_km`	Search radius in km.
`storms`	Optional character vector of storm folders; default all.
`save_dir`	Optional folder; if given, each storm's envelope, track and site series are written there as CSV, plus the event table.
`site_name`	Label used in the saved event-table file name.
`verbose`	Print progress.

Value

A data.table, one row per storm: `storm`, `year`, `name`, `basin`, `start_utc`, `end_utc`, `covers_site`, `n_site_cells`, `peak_surge_m`, `peak_surge_ft`, `peak_time_utc`, `storm_vmax_mph_10min`, `event` and `duplicate_event`.

Examples

```
ev <- slosh_events(slosh_example_dir(), lat = 29.31, lon = -94.79)
ev
```

`slosh_example_dir` *Path to the example data shipped with sloshR*

Description

Two synthetic storms (`2098Sample_tst`, `2099Demo_tst`) in the same format `slosh_convert` writes, on a small made-up grid. They exist to run the examples and tests; they are not real storms.

Usage

```
slosh_example_dir()
```

Value

Path to the example `rex_txt` folder.

Examples

```
slosh_list_storms(slosh_example_dir())
```

`slosh_footprint` *Flood footprint of one storm*

Description

Number of grid cells and approximate area (km2) where the storm's maximum surge reached each threshold. Cell areas are estimated from the spacing to neighbouring cells, so areas are approximate.

Usage

```
slosh_footprint(envelope, thresholds = c(1, 2, 3))
```

Arguments

`envelope`	Output of `slosh_envelope`.
`thresholds`	Surge thresholds in metres.

Value

A data.table, one row per threshold: `storm`, `threshold_m`, `n_cells`, `area_km2`.

Examples

```
env <- slosh_envelope(slosh_read_frames("2099Demo_tst", slosh_example_dir()))
slosh_footprint(env, thresholds = c(0.5, 1, 2, 3))
```

`slosh_list_storms` *List converted storms*

Description

List converted storms

Usage

```
slosh_list_storms(txt_dir, pattern = "*")
```

Arguments

`txt_dir`	Folder written by `slosh_convert`.
`pattern`	Optional glob pattern, e.g. `"2005*"`.

Value

A data.table: `storm`, `year`, `name` (NA when the file name has none), `basin` and `n_frames`.

Examples

```
slosh_list_storms(slosh_example_dir())
```

`slosh_plot_grids` *Plot how many storms each SLOSH grid has*

Description

Plot how many storms each SLOSH grid has

Usage

```
slosh_plot_grids(events, needed = 30)
```

Arguments

events	Output of slosh_events (all storms).
needed	Reference line for the number of events a frequency analysis typically needs. Use NULL for none.

Value

A ggplot object.

Examples

```
ev <- slosh_events(slosh_example_dir(), lat = 29.31, lon = -94.79,
                   verbose = FALSE)
slosh_plot_grids(ev)
```

---

slosh_plot_map *Map of a storm's maximum surge*

Description

Map of a storm's maximum surge

Usage

```
slosh_plot_map(envelope, track = NULL, site = NULL, title = NULL)
```

Arguments

envelope	Output of slosh_envelope.
track	Optional storm track (slosh_read_track).
site	Optional list(name = , lat = , lon =) to mark.
title	Plot title; default uses the storm name.

Value

A ggplot object.

Examples

```
d <- slosh_example_dir()
env <- slosh_envelope(slosh_read_frames("2099Demo_tst", d))
slosh_plot_map(env, slosh_read_track("2099Demo_tst", d),
               site = list(name = "Demo site", lat = 29.31, lon = -94.79))
```

slosh_plot_peaks *Plot peak surge by year*

Description

Plot peak surge by year

Usage

```
slosh_plot_peaks(events, site_name = "site")
```

Arguments

events	Output of slosh_events.
site_name	Site name for the title.

Value

A ggplot object.

Examples

```
ev <- slosh_events(slosh_example_dir(), lat = 29.31, lon = -94.79,
                   verbose = FALSE)
slosh_plot_peaks(ev, "Demo site")
```

slosh_plot_series *Plot site time series, one panel per storm*

Description

Plot site time series, one panel per storm

Usage

```
slosh_plot_series(series, site_name = "site", align_peak = TRUE, ncol = 2)
```

Arguments

series	Site series (slosh_site_series) for one or more storms stacked with rbind.
site_name	Site name for the title.
align_peak	If TRUE (default), the x axis is hours from each storm's peak; otherwise clock time (UTC).
ncol	Number of panel columns.

Value

A ggplot object.

Examples

```
d  <- slosh_example_dir()
ts <- rbind(
  slosh_site_series(slosh_read_frames("2098Sample_tst", d), 29.31, -94.79),
  slosh_site_series(slosh_read_frames("2099Demo_tst", d), 29.31, -94.79))
slosh_plot_series(ts, "Demo site")
```

---

slosh_read_frames *Read every surge frame of one storm*

Description

Read every surge frame of one storm

Usage

```
slosh_read_frames(storm, txt_dir, frames = NULL, dry_flag = 99)
```

Arguments

storm	Storm folder name, e.g. "2008Ike_gl2".
txt_dir	Folder written by slosh_convert.
frames	Optional integer vector of frame numbers to read.
dry_flag	Values with absolute value at or above this are treated as dry or missing (SLOSH writes 99.9). Default 99.

Value

A long data.table, one row per cell and frame: storm, frame, time (UTC), cell (row number in the file), lon, lat, surge_ft and surge_m (NA = dry). Heights are relative to the SLOSH basin's vertical datum.

Examples

```
fr <- slosh_read_frames("2099Demo_tst", slosh_example_dir())
head(fr)
```

slosh_read_log *Read the conversion log*

Description

Read the conversion log

Usage

```
slosh_read_log(txt_dir)
```

Arguments

txt_dir	Folder written by slosh_convert.

Value

A data.table with one row per rex file: rexfile, basin (NOAA basin code, prefixed e or h for elliptical or hyperbolic grids), imax, jmax (grid size), frames and status ("ok" or the error).

Examples

```
slosh_read_log(slosh_example_dir())
```

slosh_read_track *Read the storm track of one storm*

Description

Read the storm track of one storm

Usage

```
slosh_read_track(storm, txt_dir)
```

Arguments

storm	Storm folder name, e.g. "2008Ike_gl2".
txt_dir	Folder written by slosh_convert.

Value

A data.table, one row per frame: storm, frame, time (UTC), lat, lon (storm centre), rmax_mi (radius of maximum winds, statute miles) and vmax_mph_10min (maximum 10-minute wind, mph). Empty if the storm has no .wnd files.

Examples

```
slosh_read_track("2099Demo_tst", slosh_example_dir())
```

---

slosh_site_cells *Grid cells near a site*

Description

Grid cells near a site

Usage

```
slosh_site_cells(x, lat, lon, radius_km = 3)
```

Arguments

x	Frames (slosh_read_frames) or an envelope (slosh_envelope); needs cell, lon, lat.
lat, lon	Site latitude and longitude (decimal degrees, west negative).
radius_km	Search radius in km.

Value

A data.table of the cells within radius_km: cell, lon, lat, dist_km, nearest first.

Examples

```
fr <- slosh_read_frames("2099Demo_tst", slosh_example_dir())
slosh_site_cells(fr, lat = 29.31, lon = -94.79, radius_km = 3)
```

---

slosh_site_series *Surge time series at a site*

Description

At each time step, the highest surge among wet cells within radius_km of the site. Using several cells, not only the nearest, avoids picking a cell that is dry land in the model grid.

Usage

```
slosh_site_series(frames, lat, lon, radius_km = 3)
```

Arguments

frames	Output of slosh_read_frames.
lat, lon	Site latitude and longitude (decimal degrees, west negative).
radius_km	Search radius in km.

Value

A data.table: storm, frame, time (UTC), surge_m, surge_ft (NA = all site cells dry). Zero rows if the storm's grid does not reach the site.

Examples

```
fr <- slosh_read_frames("2099Demo_tst", slosh_example_dir())
ts <- slosh_site_series(fr, lat = 29.31, lon = -94.79)
ts[which.max(surge_m)]
```

---

slosh_storm_metrics

Storm metrics from a site time series

Description

Peak, timing, duration above thresholds and the fastest rate of rise, for one or several storms.

Usage

```
slosh_storm_metrics(series, thresholds = c(1, 2, 3))
```

Arguments

series	Output of slosh_site_series, or several stacked with rbind; needs time and surge_m, and storm to separate storms.
thresholds	Surge thresholds in metres for the duration columns.

Value

A data.table, one row per storm: storm, peak_m, peak_time (UTC), step_h (time step, hours), hours_wet (hours with any water at the site), one hours_above_<t>m column per threshold, max_rise_m_per_h and rise_h (hours from first wet step to peak).

Examples

```
fr <- slosh_read_frames("2099Demo_tst", slosh_example_dir())
ts <- slosh_site_series(fr, lat = 29.31, lon = -94.79)
slosh_storm_metrics(ts, thresholds = c(0.5, 1, 2))
```

slosh_tide_timing *Timing of the surge peak against the astronomical tide*

Description

For each storm, finds the high tide nearest the surge peak, the lag between them and the tide level at the moment of the peak. A storm that peaks near high tide produces a higher total water level.

Usage

```
slosh_tide_timing(series, tide)
```

Arguments

series	Site time series (slosh_site_series), one or several storms; needs time, surge_m (and storm).
tide	Tide predictions at the same site: a data frame with time (POSIXct or text, UTC) and level (metres). For US stations these can be downloaded from NOAA CO-OPS.

Value

A data.table, one row per storm: storm, peak_time, peak_m, tide_at_peak_m, total_at_peak_m (surge + tide), nearest_high_tide, lag_to_high_tide_h (positive = surge peak after high tide; NA when the tide series has no high water within 13 h of the peak) and max_tide_12h_m (highest tide within 12 h of the peak).

Examples

```
fr <- slosh_read_frames("2099Demo_tst", slosh_example_dir())
ts <- slosh_site_series(fr, lat = 29.31, lon = -94.79)
t0 <- min(ts$time) - 86400
tide <- data.frame(time = t0 + seq(0, 3 * 86400, by = 600))
tide$level <- 0.4 * cos(2 * pi * as.numeric(tide$time - t0) / (12.42 * 3600))
slosh_tide_timing(ts, tide)
```

slosh_transfer *Surge at other locations relative to a reference gauge*

Description

For every storm whose grid covers both the gauge and a target location, the ratio of the target's peak surge to the gauge's peak surge. A stable ratio across storms lets a long gauge-based frequency analysis be carried to locations without a gauge (a hybrid of statistics and SLOSH physics).

Usage

```
slosh_transfer(txt_dir, gauge, targets, radius_km = 3, storms = NULL)
```

Arguments

txt_dir	Folder written by slosh_convert.
gauge	list(lat = , lon =) of the reference gauge.
targets	Data frame with target (a name), lat, lon.
radius_km	Search radius around each point.
storms	Optional storm folders; default all.

Value

A list of class slosh_transfer with table (one row per storm and target: gauge_peak_m, target_peak_m, ratio) and summary (per target: n_storms, median_ratio, q25, q75, min_ratio, max_ratio).

Examples

```
tg <- data.frame(target = c("Bay head", "Open coast"),
                 lat = c(29.36, 29.26), lon = c(-94.86, -94.74))
slosh_transfer(slosh_example_dir(), gauge = list(lat = 29.31, lon = -94.79),
               targets = tg)
```

---

slosh_validate *Compare SLOSH peaks with observations*

Description

Matches modelled and observed peaks by storm and reports the error of each storm and summary skill scores.

Usage

```
slosh_validate(sim, obs)
```

Arguments

sim	Modelled peaks, usually slosh_events: needs storm and peak_surge_m; peak_time_utc optional.
obs	Observed peaks: needs storm and obs_peak_m; obs_peak_time (UTC) optional. Observed and modelled heights must share a datum (convert first if not).

Value

A list of class `slosh_validation` with `table` (one row per matched storm: modelled and observed peak, `error_m` = model minus observation, and `timing_error_h` when times are given) and `summary` (`n`, `bias_m`, `mae_m`, `rmse_m`, `r`, `mean_timing_error_h`).

Examples

```
ev  <- slosh_events(slosh_example_dir(), lat = 29.31, lon = -94.79,
                    verbose = FALSE)
obs <- data.frame(storm = ev$storm, obs_peak_m = ev$peak_surge_m + c(0.2, -0.1))
slosh_validate(ev, obs)
```

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